TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Fabric and Garment CAD Software

Computer-aided design for the two drawing offices of the textile trade: fabric CAD, where weaves, knits and colourways are built and previewed on screen, and garment CAD, where patterns are drafted, graded and laid into markers without paper, pins or a single cut piece of cloth.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 3 · LEVEL 3 TERM 2

1. Aim

To work through a complete garment-CAD task sequence — drafting a basic bodice block on screen, adding seam allowances and notches, grading it to a size range and checking a marker's efficiency — and to state what fabric CAD contributes on the cloth side of the same style.

2. Concepts you need

- **Garment CAD (pattern design system)** covers four linked jobs: **drafting/digitising** (creating pattern pieces as vector outlines — points, lines and curves with exact coordinates), **modification** (moving points, adding darts, seam allowances and notches), **grading** (growing the base size into the size range by rule-based point movements), and **marker making** (nesting all pieces of all sizes onto the fabric width to minimise waste).
- **Vector, not picture:** a CAD pattern is geometry, not an image. Every point has coordinates, so the system can report any measurement (chest, waist, sleeve length) exactly, check that mating seams are equal, and drive a plotter or an automatic cutter. Zooming never loses accuracy.
- **Grading rules:** each pattern point carries a grade rule — how far it moves in X and Y per size step. The computer applies the rules; the pattern master remains responsible for the rules themselves (e.g. chest +5 cm per size, length +1 cm).
- **Marker efficiency** is the percentage of the marker's cloth area actually covered by pattern pieces: *efficiency = (total piece area ÷ (marker length × fabric width)) × 100*. It is the single number that decides whether a marker is good — a 2 % gain on a large order is tonnes of fabric.
- **Fabric CAD** works one stage earlier: woven designs are built from weave notation, yarn counts and colour order, then rendered as a realistic cloth simulation; knit structures and printed designs are

handled the same way. The buyer approves a screen simulation or a small strike-off instead of a full sample loom run.

- **The digital thread:** the graded pattern file feeds the marker, the marker feeds the cutter, and the same style file later feeds 3D simulation (Experiment 4) — one source of geometry for the whole sample room.

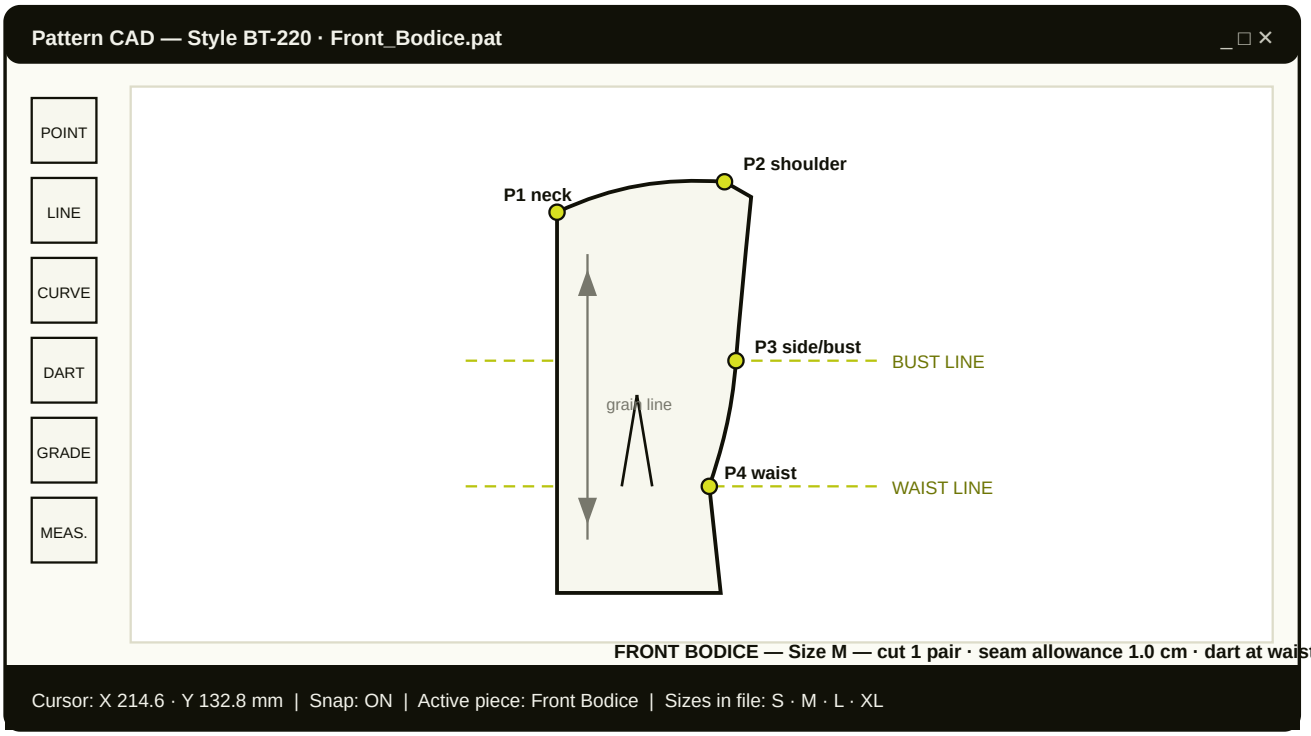


Figure 1 — The pattern-CAD working screen in schematic form: tool palette at the left, the vector pattern piece with its control points (P1–P4), construction lines and grain line in the canvas, and the coordinate/status bar below. Every point shown is a coordinate pair the system can measure and grade.

3. Lab procedure — the task sequence

1. **Set up the size chart:** enter the base size M measurements (chest 107 cm, waist 82 cm, body length 72 cm) and the grade rules (chest ±5 cm, waist ±4 cm, length ±1 cm per size).
2. **Draft the block:** construct the front bodice from the measurement table — plot P1 (neck), P2 (shoulder), P3 (side at bust), P4 (waist); draw the neckline and armhole as curves through their control points; close the outline and let the system verify the shape is closed.
3. **Add engineering detail:** waist dart, notches at the armhole and side seam, grain line, piece name/quantity text, and 1.0 cm seam allowance all round (2.5 cm at the hem).
4. **Check:** use the measure tool on bust and waist; walk the front and back side seams against each other — the system reports any length mismatch. Correct until mating seams agree.

5. **Grade:** apply the grade rules to produce S, M, L, XL; display the graded nest and inspect that curves stay smooth and darts grade proportionally.
6. **Make a marker:** place all pieces of the style on 150 cm fabric width; record the marker length and the efficiency the system reports; rearrange the two worst-fitting pieces and record the improvement.
7. **Output:** save the style file under the naming rule and plot/print the base-size piece at full scale; verify one plotted measurement against the size chart with a tape.

4. Worked example — grading and marker efficiency

Grading: with the chest grade of 5 cm per size applied at the bust points of all pieces, the size range of the style is:

$$\text{Chest: S} = 107 - 5 = \mathbf{102\ cm} \cdot \text{M} = \mathbf{107\ cm} \cdot \text{L} = 107 + 5 = \mathbf{112\ cm} \cdot \text{XL} = \mathbf{117\ cm}$$

Marker efficiency: the marker for the full style (front, back, sleeves, collar) comes out at 6.20 m long on 1.50 m wide fabric, and the system reports a total piece area of 7.90 m²:

$$\text{Marker area} = 6.20 \times 1.50 = 9.30\ \text{m}^2 \ \rightarrow\ \text{Efficiency} = (7.90 \div 9.30) \times 100 = \mathbf{84.9\ \%}$$

The remaining 15.1 % is cutting waste between pieces — the figure the marker maker is paid to shrink, and the figure used by merchandising when the fabric consumption per garment is costed. Note that consumption follows directly: 9.30 m² of marker ÷ 6 garments per marker ≈ 1.55 m² of fabric bought per garment at this efficiency.

5. Practice tasks

1. A marker is 5.80 m long on fabric 1.45 m wide; total piece area is 7.15 m². Calculate the marker efficiency, and state whether it is better or worse than the worked example.
2. Using a waist grade of 4 cm per size from a base M of 82 cm, give the waist measurements of S, L and XL.
3. A digitised paper pattern fails the system's "closed shape" check at one corner. State what this means physically, what happens if it is ignored and sent to the cutter, and the correct fix.

Answer key

1. Marker area = 5.80 × 1.45 = 8.41 m². Efficiency = (7.15 ÷ 8.41) × 100 = **85.0 %** — marginally **better** than the worked example's 84.9 %.
2. S = 82 − 4 = **78 cm**; L = 82 + 4 = **86 cm**; XL = 82 + 8 = **90 cm**.
3. The outline's end point does not meet its start point — there is a gap, so the "piece" is an open chain, not a closed area: the system cannot compute its area, place seam allowance around it reliably, or

nest it correctly. Sent to an automatic cutter, an open contour produces a piece cut with a slit or an incomplete edge — scrap at fabric prices. The fix is in the CAD file: zoom to the corner, snap/join the endpoints (using the point-join or close-shape tool), and re-run the closed-shape and seam-walk checks before saving.
